

Minimum Essential Characteristics (MEC)

Serological Automated Analyzers

Serological Automated Analyzers:

- A. Specimen and reagent mechanisms that involve the use of a single independent processing system for each unique specimen and reagent; thus, preventing cross contamination between specimens and/or reagent reservoirs. The mechanism may include a multiple channel but shall be an independent process system.
- B. Sampling system which auto-detects specimen volume integrity as well as loss of integrity of the pressure gradient for any independent processing unit.
- C. Shall be capable of performing high-volume testing with minimal personnel intervention (staffing must be able to load analyzer, then walk away). To enable workflow efficiency and minimal processor staffing initial run set-up must be equal to or greater than 700 samples per analyzer with the ability to continuously load specimens and walk away. Shall have capability to perform up to 1,600 samples within an 8 hour period, 6 days per week.
- D. Shall have the capability to perform testing from an 8.5 mL 16x100 mm serum separation tube (SST).
- E. Shall have the capability of performing testing on grade 4+ (gross) hemolytic specimens.
- F. Shall employ fully automated, robotic processing from addition of diluents, samples, incubation, wash steps, and calculation of results with limited operator interaction.
- G. Shall have the capability to interface with a variety of Laboratory Information Systems (LIS) such as Labware, and Laboratory Information Management System (LIMS).
- H. Power supply shall include conditioner power source which will help to eliminate the fluctuation in power going to the instrument especially during a power failure which could affect the operation of the instrument.
- I. Shall not require plumbing or a drainage system or external ventilation of analyzers.
- J. Shall be a U.S. Food and Drug Administration (FDA) approved as Class III in vitro diagnostic device or instrument for conduct of an FDA approved 4th Generation HIV-1/HIV-2 screening test.
- K. Service agreement shall be standard, 8 hours/day, 5 days/week, as well as bi-annual preventive maintenance.

Confirmation Analyzers:

- A. Human immunodeficiency virus (HIV) confirmation analyzer shall be compatible with the Western Blot test system.

- B. HIV confirmation analyzer shall have the ability to detect the separate bands of the HIV-1 and HIV-2 virus.
- C. Shall be FDA approved for use in confirmation testing for the presence of the HIV virus.

Reagents:

- A. Reagents (assay) shall be capable of being run as comparator, anti-HIV-1 antibody assay in a manual test format, or on an open analyzer system.
- B. Combo HIV-1/HIV-2 reagents shall have the ability to detect HIV p24 antigen as well as the presence of HIV antibodies.
- C. Combo HIV-1/HIV-2 reagent sensitivity shall have the ability to detect the presence of HIV antigen/antibodies within 7-10 days of infection.
- D. Shall be capable of shipping reagents on minimum purchase order to cover 10,000 screening tests and maximum purchase order of 48,000 screening tests. The kit test size shall be the most appropriate size in order to decrease the storage space required.

HIV Confirmation Reagents:

- A. Shall have the ability to differentiate between HIV-1 and HIV-2.
- B. Shall be compatible with the Western Blot System.
- C. Shall be capable of detecting the separate bands of the HIV Virus.
- D. Shall have capability to meet the maximum purchase order to cover 500 HIV confirmation tests and minimum purchase order of 200 HIV confirmation tests. The kit test size will be left to the discretion of the vendor.